

Pathology of Bone and Joints – Part 2

Principles and Practices of Osteopathic Medicine I

Eugen Petcu MD, PhD

**Associate Professor
Department of Biomedical
Sciences
epetcu@nyit.edu**

Do.
Make.
Heal.

Innovate.

Reinvent the Future.

OUTLINE

- Fractures
- Osteomyelitis
- Seronegative Spondyloarthropathies (ankylosing spondylitis, reactive arthritis, infectious arthritis), Lyme arthritis
- Fibrous Dysplasia, Simple Bone Cyst, Aneurysmal bone cyst
- Bone Tumors Overview
- Benign and Malignant Tumors of Bone and Cartilage

FRACTURES

Loss of bone integrity

Classification

Simple: overlying skin intact

Compound: bone communicates with skin surface

Comminuted: bone fragmented

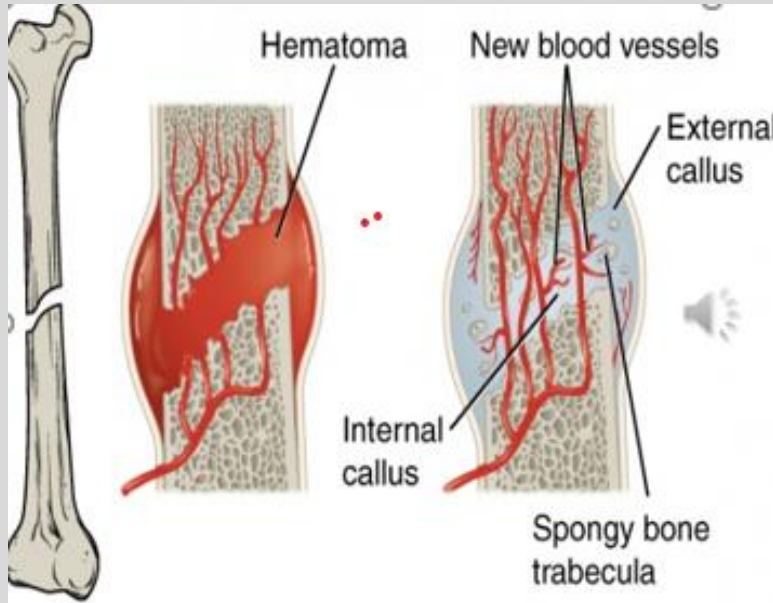
Displaced: ends of bone at fracture site not aligned

Stress: due to increased physical activity; the bone is damaged by repetitive loads

Greenstick: in infants (bone are soft), extend only partially through the bone

Pathologic: disease process (e.g., tumor)

FRACTURE REPAIR



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Repair of bone : genetically mediated

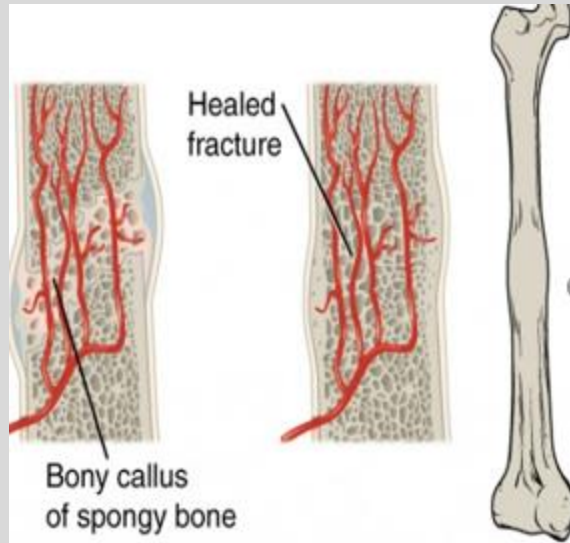
Immediately after fracture: **Hematoma** formation → provides fibrin mesh (fracture sealing) → inflammatory cell influx, fibroblast ingrowth, and capillary proliferation

→ **Granulation Tissue**

Degranulated platelets and inflammatory cells → PDGF, TGF- β , FGF → activate osteoprogenitor cells in periosteum, medullary cavity, and surrounding soft tissues stimulating **osteoclastic and osteoblastic activity** →

Soft Tissue Internal Callus or Procallus, provides some anchorage but not structural rigidity for weight bearing

FRACTURE REPAIR (continued)



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Within 2 weeks of injury, activated osteoprogenitor cells : subperiosteal trabeculae of woven bone oriented perpendicular to cortical axis and within medullary cavity:

➡ **Procallus transforms into bony callus** (max girth at end of 2nd or 3rd week); fx stabilization

Endochondral ossification: contiguous network of *bone and newly deposited bone trabeculae* (medulla, beneath periosteum); fx bone **ends bridged**; progressive mineralization, stiffness and strength of callus increases weight bearing

Healing process completed by restoration of medullary cavity

Inadequate immobilization: movement of the callus ➡ delayed union or nonunion

If nonunion persists: the malformed callus undergoes **cystic degeneration**; luminal surface can become lined by synovial-like cells, creating **false joint or pseudoarthrosis**

OSTEOMYELITIS

Inflammation of bone and marrow, secondary to bacterial, viral, fungal or parasitic infection.

No specific organism is identified in nearly 50% of patients.

Pyogenic osteomyelitis - bacterial infection:

- hematogenous spread
- extension from a contiguous site
- direct implantation (e.g., in traumatic injury)

Staphylococcus aureus: 80% to 90% of culture-positive pyogenic osteomyelitis

Escherichia coli , **Pseudomonas** , and **Klebsiella**

Haemophilus influenzae and **group B streptococci** : neonates

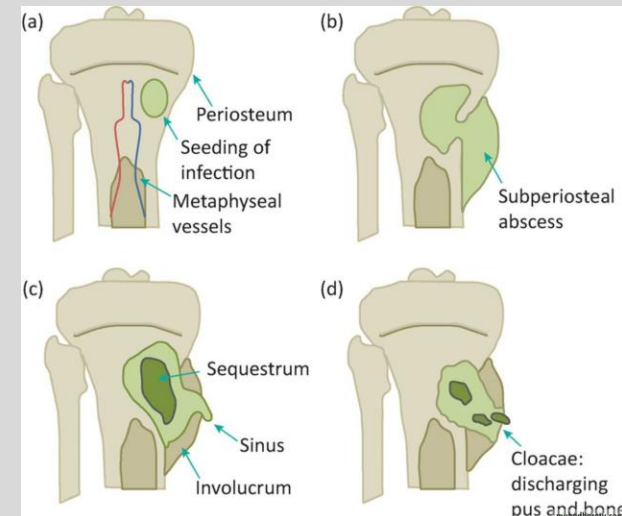
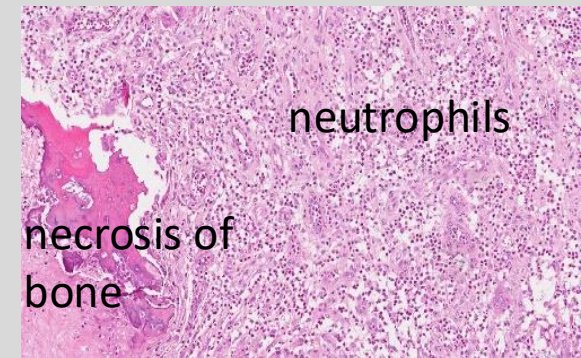
Sickle cell disease predisposes to Salmonella infection

OSTEOMYELITIS

- **ACUTE:** bacteria proliferate and recruit neutrophils
within 48 hours: necrosis of bone cells and marrow
bacteria: periosteum → **subperiosteal abscess**
- → **periosteal lifting further impairs blood supply**
- → **necrosis**
- → soft tissue abscesses after periosteal rupture – open to skin as **draining sinuses**
- Dead bone, or **sequestrum** is formed and this structure can crumble and release fragments into sinus tract forming a so called CLOACAE indicating a **CHRONIC** Stage; this is seen in 5% to 25% of cases
- As inflammation continues, chronic inflammatory cells recruited during first week release cytokines that stimulate bone resorption, fibrous tissue ingrowth, and peripheral deposition of reactive bone; **this new bone can form a living shell, or involucrum, around devitalized, infected bone**
- **Xray:** lytic focus of bone destruction surrounded by sclerosis

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Draining osteomyelitis:

- Drainage tract in the subperiosteal shell of viable new bone (**involucrum**) reveals inner native necrotic cortex (**sequestrum**)



SERONEGATIVE SPONDYLOARTHROPATHIES

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Undefined microbial antigens  cross-react with musculoskeletal
system components  T-cell responses 
Immune Mediated Seronegative Spondyloarthropaty

Heterogeneous group of disorders

- Absence of rheumatoid factor
- Pathologic changes in ligamentous attachments rather than synovium
- Sacroiliac joint involvement
- Association with **HLA-B27**
- Bony proliferation leading to ankylosis

Examples: Ankylosing spondylitis, infectious arthritis, reactive arthritis

ANKYLOSING SPONDYLITIS:

Destruction of articular cartilage and bony ankyloses:

- sacroiliac and vertebral apophyseal joints between tuberosities and processes

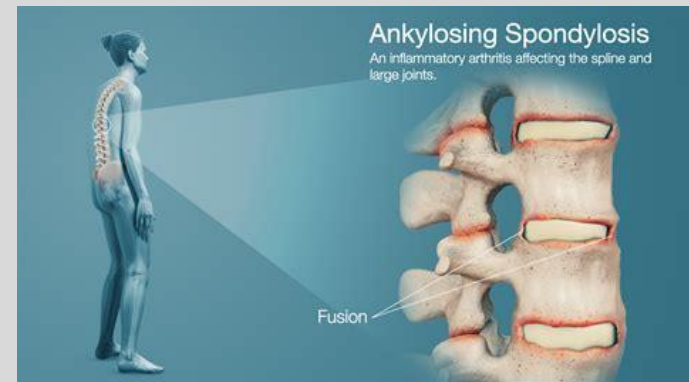
Clinical: **lower back pain and spinal immobility (2nd and 3rd decades of life)**

- Approximately 90% of patients HLA-B27 positive; role unknown

INFECTIOUS ARTHRITIS

Microorganisms:

- hematogenous dissemination
- direct inoculation
- contiguous spread from soft tissue (abscess, osteomyelitis)



REACTIVE ARTHRITIS (REITER'S SYNDROME)

TRIAD: arthritis, nongonococcal urethritis or cervicitis, and conjunctivitis

- Mono- or oligoarticular arthritis
- Intestinal infections: Salmonella, Shigella, Campylobacter
- STD: Chlamydia trachomatis
- Rarely: group A streptococcal infections
- RF negative; HLA-B27 positive
- young adults
- 50%: Recurrent Arthritis, Tendonitis, Lumbosacral pain;
- Active disease > 6 months: **CHRONIC REACTIVE ARTHRITIS**



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Presentation:

Initial acute-onset asymmetric oligo-arthritis (knee) often with inflammatory low back pain. In some patients: Inflammation at insertions of Achilles tendon and plantar fascia with heel swelling; inflammation of digits (dactylitis), joint stiffness, and low back pain.

Extraarticular :

- conjunctivitis and uveitis;
- dysuria, pelvic pain, urethritis,
- balanitis, cervicitis, mucosal ulcers,
- skin rashes, and cardiac valvular disease

LYME ARTHRITIS

Agent: Spirochete *Borrelia burgdorferi*

Deer ticks (*Ixodes*);

Leading arthropod-borne disease in US

New England, mid-Atlantic states, and upper Midwest

Classic Lyme disease – involves multiple organ systems; and has three clinical phases:

Early localized stage: initial skin infection, followed (days/weeks) by dissemination of organism to other cutaneous sites, cranial nerves, heart, and meninges

➡ **Early disseminated stage**

The untreated **or the late disseminated disease** is associated with a migratory arthritis (**Lyme arthritis**) in 80 % of cases (**months after infection, lasts weeks to months**)

Arthritis: 10% of *Borrelia* infections

Most are treated and cured at earlier stage

Joints: knees, shoulders, elbows, and ankles
(one or two joints)

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- **Rash- Lyme rash:** bull's-eye with circles around the middle. However, most are round, red, and at least 2 inches across.
- Grows to 10- 12 inches across within days.
- Warm to the touch
- Not itchy or pain
- Location: anywhere

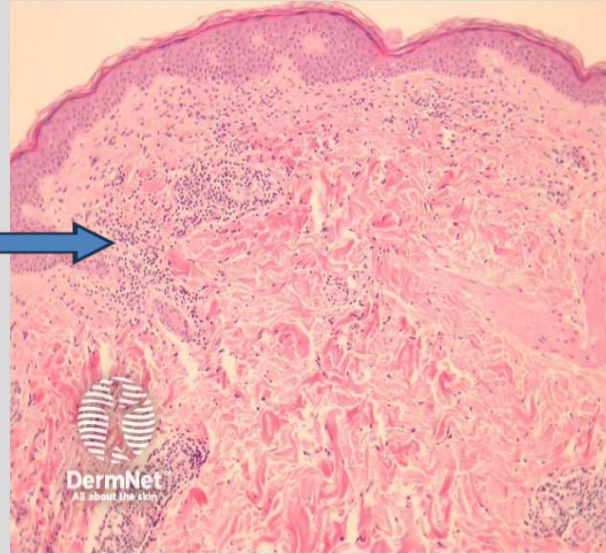


Lyme rash

Lyme arthritis

Biopsy of the bull's eye rash: sparse superficial and deep lymphocytic infiltrate

under an unremarkable epidermis
+/- Epidermal and/or dermal necrosis.



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An infected synovium shows chronic synovitis, synoviocyte hyperplasia, fibrin deposition, CD4+ T cells infiltrates, and obliterative endarteritis

Serology: anti- Borrelia antibodies

Treatment: doxycycline, amoxicillin, or cefuroxime axetil; cure rates of 90%

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JOHNSTON ET AL

AJP • January 1991

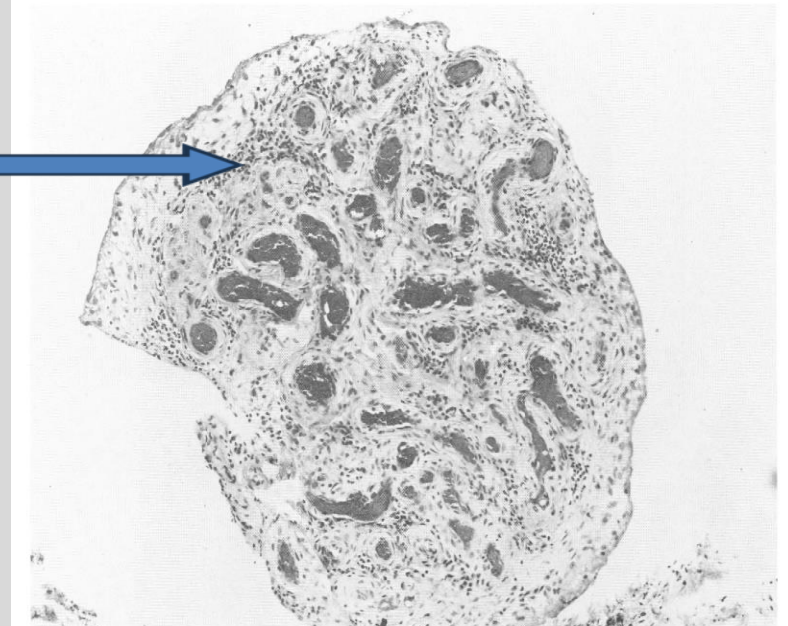


Figure 2—Cross-section of a Lyme arthritis synovial villus with accentuated and congested microvasculature. (H&E, x 40)

FIBROUS DYSPLASIA

Benign tumor: all components of normal bone present → do not differentiate into mature structures;

During skeletal development .

Pathogenesis: various somatic gain-of-function mutations in GNAS1 (encodes the stimulatory α subunit of G s, resulting in constitutively active G s protein) → promoting cellular proliferation and inhibition of osteoblast differentiation.

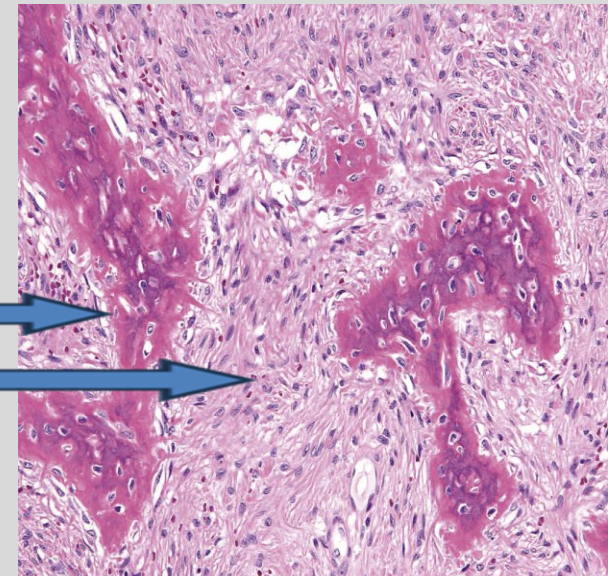
Intramedullary lytic lesions: bowing and cortical Bone thinning;

Histo:

curvilinear trabeculae of woven bone →
moderately cellular fibroblastic proliferation →
without prominent osteoblastic rimming

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FIBROUS DYSPLASIA

Clinical: pain, swelling, pathologic fractures, severe bone deformity

Monostotic fibrous dysplasia - early adolescence: femur, tibia, ribs, jawbones, and calvarium usually; asymptomatic, can cause pain, discrepancies in limb length, and pathologic fracture; treated by curettage, recurrence common

Polyostotic fibrous dysplasia - earlier age, developing into adulthood; femur, skull, and tibia; craniofacial involvement; progressive with crippling deformities, and fractures; bisphosphonates to reduce severity of bone pain; **rare malignant transformation into sarcoma**

McCune-Albright syndrome — precocious sexually development, esp. in girls; hyperthyroidism, pituitary adenomas that secrete growth hormone, and primary adrenal hyperplasia; unilateral; skin lesions (café-au-lait spot) on same side of body



SIMPLE BONE CYST (UNICAMERAL BONE CYST)

- Incidental finding
- Not a neoplasm
- Fluid filled lesion with fibrous lining
- During the first 20 years of life
- Near the growth plate of long bones
- Reoccur/disappear with maturity
- If large, may fracture



Aneurysmal Bone Cyst (ABC)

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Multiloculated blood-filled spaces.; all ages affected, most in adolescence; often in femur, tibia, vertebral body

Xray: expansive, well-circumscribed lytic lesion with well-defined margins; most show central lysis and thin sclerotic “eggshell” of reactive bone at periphery
Multiple blood-filled cystic spaces separated by thin, tan-white septa composed of plump spindle cells, multinucleated osteoclast-like giant cells, and reactive woven bone lined by osteoblasts

70%: chromosome 17p13 rearrangements present within plump spindle cells → results in fusion of **USP6* coding region** with promoters of genes highly expressed in **osteoblasts** (CDH11)

Clinical: localized pain and swelling → limp, vertebral lesions, and nerve compression

Benign, but locally aggressive
treatment by curettage or excision with recurrence in 10% to 50% of cases



Ganglion and Synovial cysts

Ganglion cysts: up to 1.5 cm, near a joint capsule or tendon sheath; joints of wrists; firm, fluctuant, pea-sized translucent nodules

Cystic or myxoid connective tissue degeneration;

Cyst wall lacks cell lining

It may be multilocular: enlarge through coalescence and degeneration of adjacent connective tissue

Synovial cyst: herniation of synovium through a joint capsule or massive enlargement of a bursa

Examples: Popliteal Synovial Cyst, or Baker cyst, associated with rheumatoid arthritis;
Cyst lining resembles synovium
Cyst and synovium :hyperplastic;
Cyst fluid: inflammatory cells and fibrin



PRIMARY BONE TUMOURS

- Rare; outnumbered by metastases;
- Long bones of extremities
- Asymptomatic, identified incidentally
- Some are painful
- Slow-growing mass
- Pathologic fracture

- Important factors
 - Clinical Presentation
 - Radiographic Appearance
 - Age of Patient
 - Location of Lesion
 - Microscopic Appearance

During the first 3 decades of life—the ages of the greatest skeletal growth activity

Location: Distal femur and proximal tibia (bones with highest growth rate)

Relatively specific radiographic presentations

Benign >> Malignant

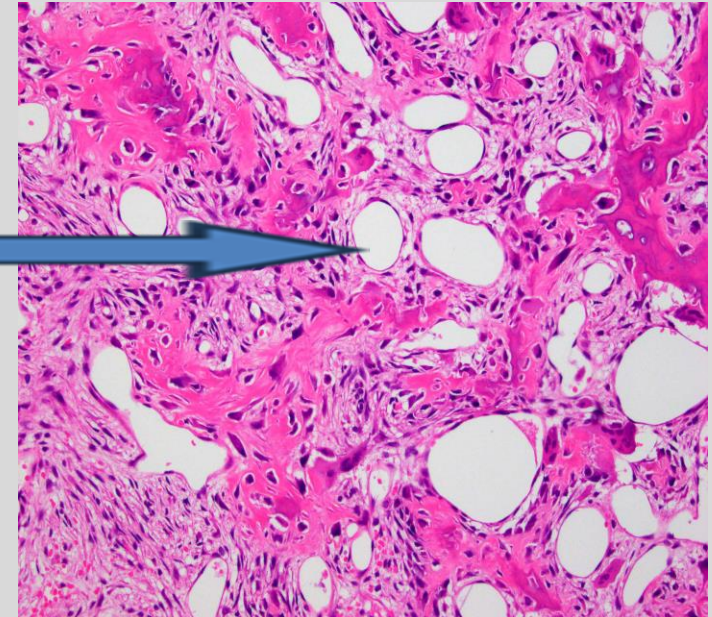
Age probably most important clinical clue to diagnosis

Painful lesions: growing lesions (malignancy or locally aggressive)

BENIGN TUMORS: OSTEOID OSTEOMA AND OSTEOLASTOMA

- Benign bone-producing tumors: **similar histologic features** but different sizes, sites of origin, and symptoms; malignant transformation rare
- Round-to-oval, well-circumscribed masses of hemorrhagic, gritty tan tissue
- **Micro:** trabeculae of woven bone rimmed by single layer of prominent osteoblasts, loose connective tissue with dilated and congested capillaries
- Reactive bone encircles actual neoplasm, or nidus (small mineralized radiolucent core)
- **Osteoid osteoma:** tx by radiofrequency ablation
- **Osteoblastoma:** curettage or en bloc excision

woven bone



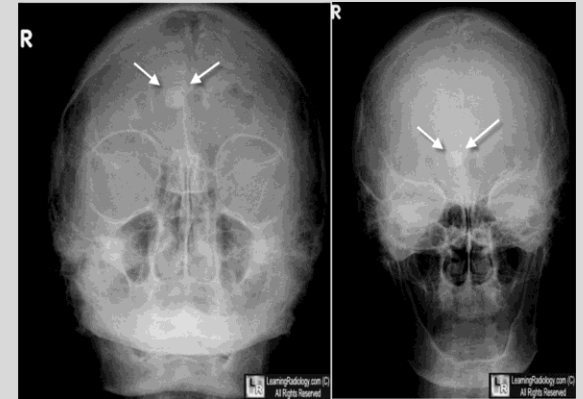
OSTEOID OSTEOMA

- Benign tumor of bone with no malignant potential
- Less than **< 2 cm in size**
- Males <25 years old; **long bones** and on the surface of facial bones or skull

Gardner's Syndrome:

Familial Adenomatous Polyposis + osteomas and soft tissue tumors

- Slow growing, no clinical significance unless : obstruction or cosmetic issues
- **Nocturnal bone pain** (secondary to excess prostaglandin E2 produced by proliferating osteoblast), that is **relieved by Aspirin and NSAIDS**
- **Clinical:** a thick rim of reactive cortical bone may be only investigational finding



OSTEOBLASTOMA



- > 2cm in size
- More common in males <25 years old
- Arises in **spine**
- Pain which is not nocturnal and it is **NOT** relieved by aspirin or other NSAIDS

OSTEOCHONDROMA (EXOSTOSIS)

Common

Attached to skeleton by **bony stalk capped by cartilage**

- 85%: solitary and sporadic,
- 15: part of the the AD multiple hereditary exostosis syndrome
- **Solitary osteochondromas**
- in late adolescence/ early adulthood
- men affected 3X more than women
- in bones of endochondral origin

Site: metaphysis near growth plate of long tubular bones (knee); palpable

Slow-growing masses

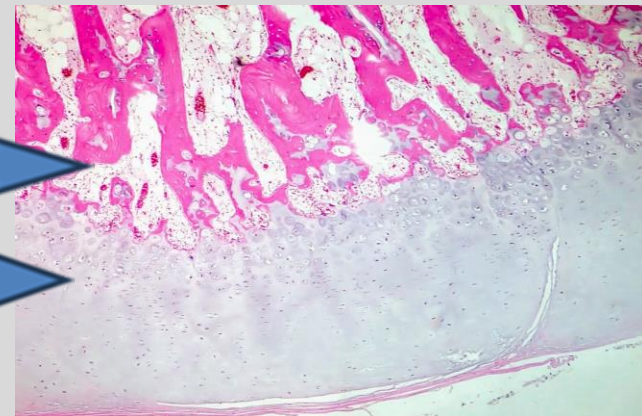
Pain: nerve impingement

Histology

- disorganized growth plate, which undergoes endochondral ossification;
- cartilage cap lined by perichondrium, contiguous with mature bone



Sessile or Pedunculated osteochondroma fragments, and range in size from 1 to 20 cm



CHONDROMA (ENCHONDROMA)

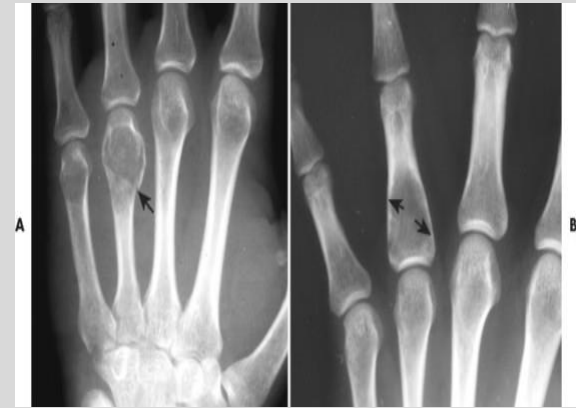
- **Benign tumor of hyaline cartilage** : within medullary cavity (enchondromas) or the on the bone surface (juxtacortical chondromas)
- The patients are 20 – 50 years old;
- Asymptomatic
- Painful; Pathological Fractures
- Well-circumscribed nodules usually < 3 cm
- Cut surface: gray-blue, and translucent
- Heterozygous mutations in IDH1 and IDH2 genes

Histology: hypocellular lesion with an abundance of hyaline cartilage. The chondrocytes appear normal. A peripheral endochondral ossification may be noted.

Xray: circumscribed lucency with central irregular calcifications, sclerotic rim, and intact cortex

Treatment: usually observation or curettage;

Solitary chondromas rarely undergo sarcomatous transformation, but those seen in Ollier disease and Maffucci syndrome do so more frequently



Ollier disease and Maffucci syndrome: nonhereditary disorders characterized by multiple enchondromas

Ollier disease: enchondromas confined to one side and limited to the limbs. Increased risk of chondrosarcoma 25-30% at 40 years . Increased risk of developing gliomas, pancreatic and juvenile granulosa cell tumors of the ovary.

Maffucci syndrome: multiple enchondromas and spindle cell hemangiomas. The patients are at risk of developing other malignancies, including brain gliomas

GIANT CELL TUMOR (Osteoclastoma)

- Rare benign tumor of multinucleated osteoclast-type giant cells; locally aggressive
- 4% metastasize to lungs or they may regress spontaneously
- rarely fatal
- knee, involving distal femur or proximal tibia
- Arthritis-like symptoms
- Pathologic fractures;
- May destroy overlying cortex, resulting in bulging soft tissue mass

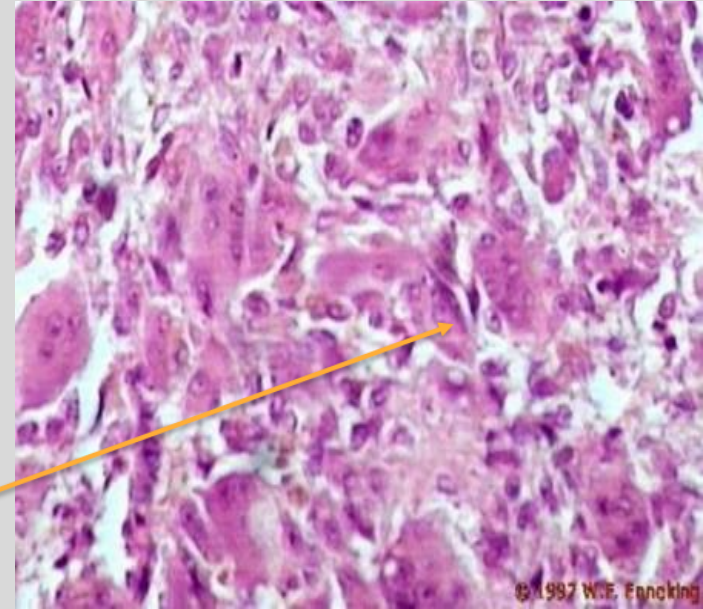
Macroscopically, these are large, red-brown masses that frequently undergo cystic degeneration

Histology shows uniform oval mononuclear tumor cells and abundant **osteoclast-type giant cells with 100 or more nuclei**;

- **Necrosis and mitotic activity may be prominent**

Xray: Soap Bubble

Therapy: curettage, 40% to 60% recur locally;



OSTEOSARCOMA

Most common primary malignant tumor of bone;
20% of bone cancers

Bimodal age distribution of osteosarcoma

75% before 20 years of age and a smaller peak in older adults:

- Paget disease
- Bone infarcts
- Prior radiation

M/F:1.6 : 1

In metaphyseal region of long bones;
50% of cases near knee in distal femur or proximal tibia

OSTEOSARCOMA MORPHOPATHOLOGY

Bulky, gritty, gray-white tumors, usually with hemorrhage and cystic degeneration; destroys surrounding cortices to produce soft tissue masses, spread extensively into medullary canal, replacing hematopoietic marrow

Histology: pleomorphism, large hyperchromatic nuclei, bizarre tumor giant cells, and abundant mitoses including abnormal (e.g., tripolar) forms; extensive necrosis and intravascular invasion common

Diagnosis: presence of malignant tumor cells producing unmineralized osteoid or mineralized bone

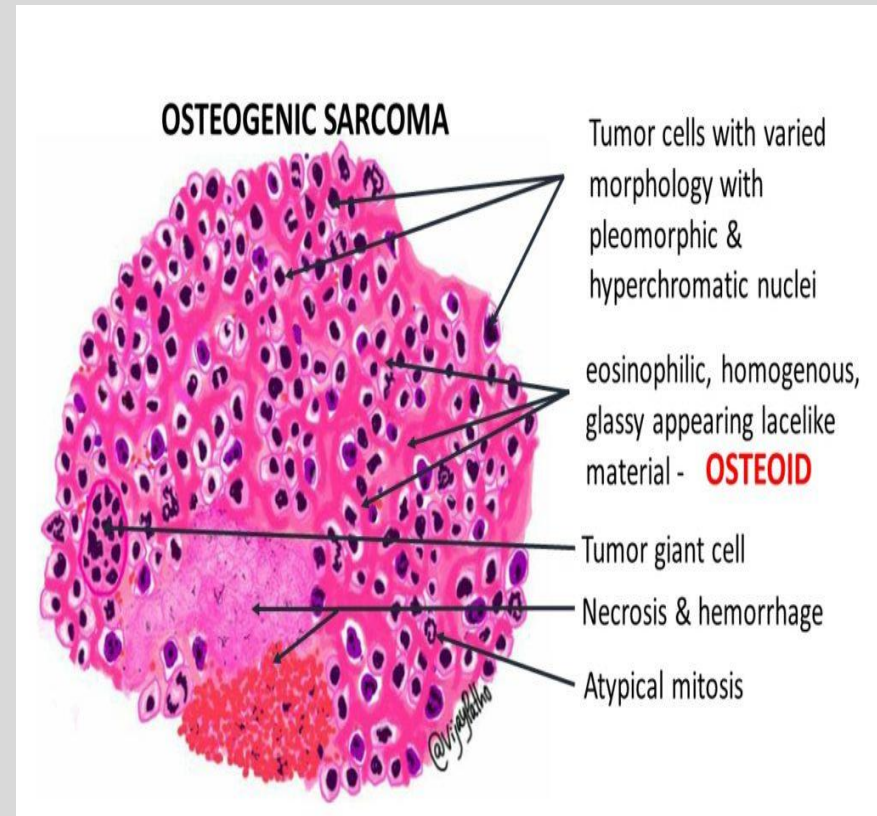
All cases: assumed to have occult metastases at time of diagnosis →

Hematogenous metastases :lungs, bones, brain, and other sites

5-year survival rate is less than 20% for those with metastases, recurrent disease, or secondary osteosarcoma

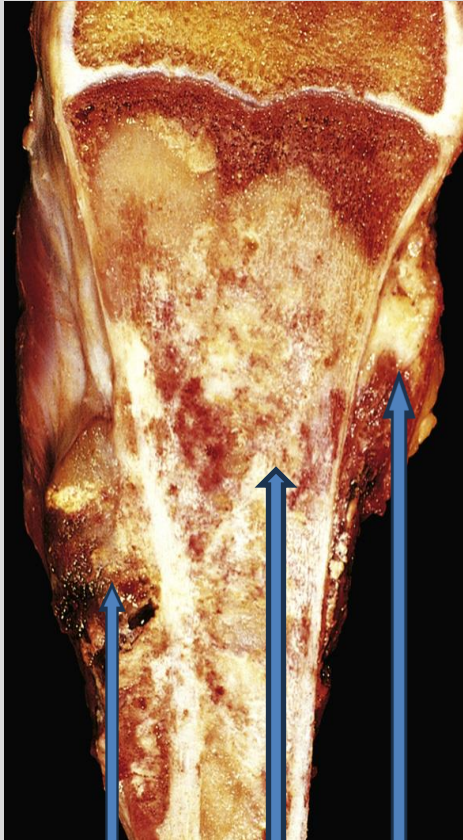
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OSTEOSARCOMA

Osteosarcoma of proximal tibia

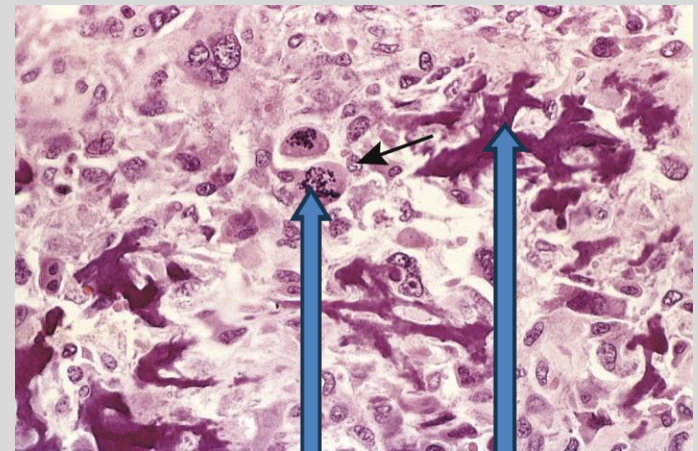


Tumor : medullary cavity of the metaphysis and proximal diaphysis.

Infiltration through the cortex, lifted the periosteum, and formed soft tissue masses on both sides of the bone

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Lacelike pattern of neoplastic bone produced by anaplastic malignant tumor cells.

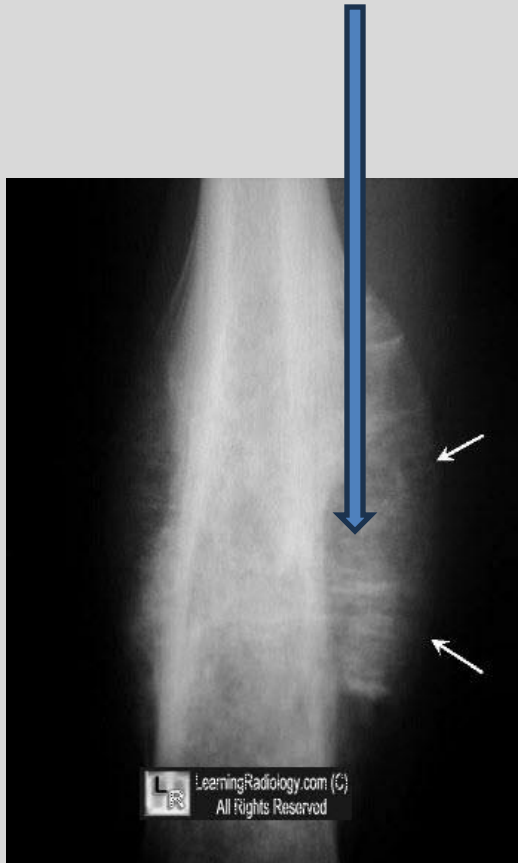
Anormal mitotic figures.

OSTEOSARCOMA CLINICAL and GENETIC FEATURES

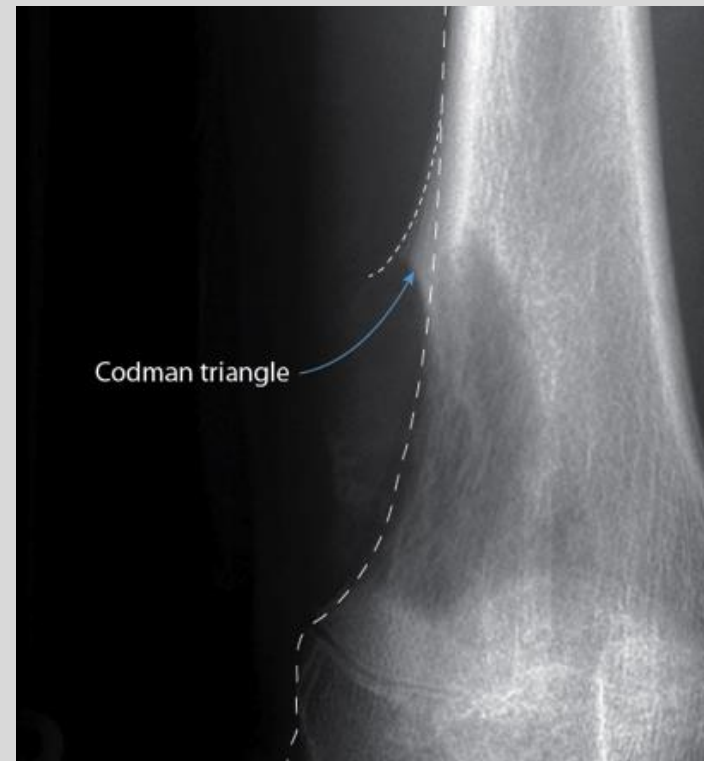
- Pain and Pathologic fractures
- Peak incidence: adolescent growth spurt;
- Location: in the growth plate of rapidly growing bones (as increased proliferation may predispose to mutations)
- Acquired genetic abnormalities including chromosomal aberrations; usually associated with mutations such as:
 - RB mutations in up to 70% of sporadic osteosarcomas;
 - Germline mutation of TP53 in individuals with Li-Fraumeni syndrome: sarcomas and (osteo)sarcoma, breast, brain, and adrenal cancer
 - CDKN2A (encodes two tumor suppressors, p16 and p14) inactivated in many osteosarcomas

OSTEOSARCOMA

- **Xray:** destructive, mixed lytic and blastic mass with infiltrative margins; often it breaks through the cortex and lifts periosteum, inducing reactive periosteal bone formation.
- **Sunburst Pattern:** stretched-out bone fibers from rapid growth



Triangular shadow between cortex and raised periosteal ends : **Codman triangle** = aggressive tumor



Ewing Sarcoma

- Malignant bone tumor: undifferentiated primitive round cells without differentiation;
- 6% to 10% of primary malignant bone tumors;
- 2nd most common group of pediatric bone sarcomas
- 80% of patients <20 years of age;
- Boys > Girls with predilection for Caucasians;
Rarely African or Asian descent

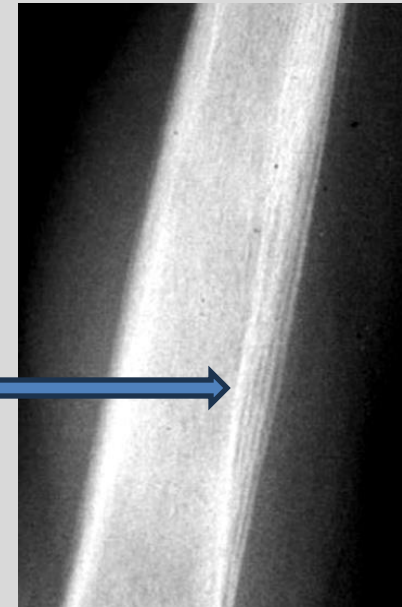
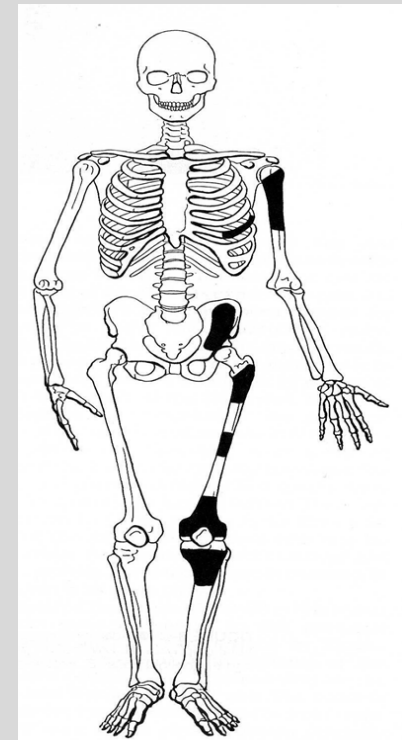
Location: Diaphysis of long tubular bones, femur and pelvis, humerus and ribs

Presentation:

- painful, enlarging mass
- affected site is frequently tender, warm, and swollen;
- systemic findings that mimic infection: fever, elevated ESR, anemia, and leukocytosis

Xray:

- lytic tumor with permeative/moth eaten margins
- periosteal reaction produces
- reactive bone deposited in an onion-skin fashion



t(11;22); > 90%: balanced translocation involving EWSR1 gene on chromosome 22 and FLI1 gene on chromosome 11



EWSR1/FLI1 fusion gene
encodes a chimeric EWS/FLI1 protein



binds to chromatin and dysregulates transcription



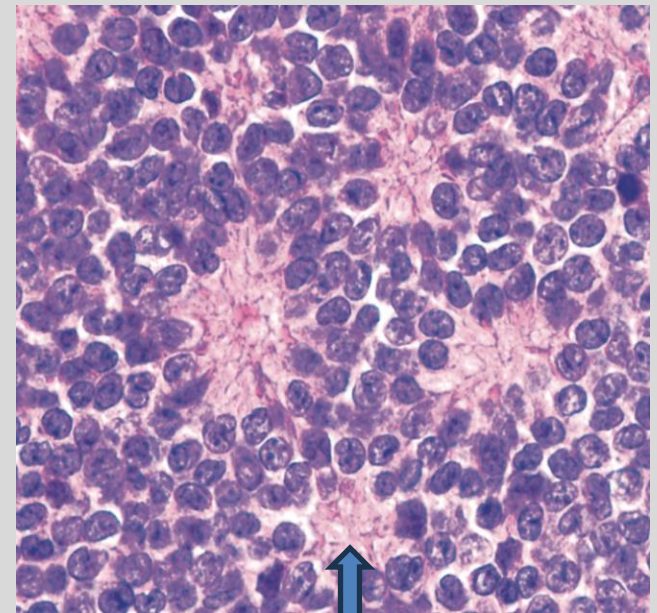
uncontrolled growth and abnormal differentiation

Origin: mesenchymal stem cells?
primitive neuroectodermal cells?

Starts in medullary cavity and invades cortex, periosteum, and soft tissue

Macro: Soft, tan-white, with areas of hemorrhage and necrosis

- Aggressive
- Neoadjuvant chemotherapy followed by surgical excision and adjuvant chemotherapy, with or without irradiation, has achieved 75% 5-year survival and 50% long-term cure rates



Micro: sheet of uniform small, round cells with scant cytoplasm

Homer-Wright rosettes: rounded cell clusters with a central fibrillary core, indicating a neuroectodermal differentiation.

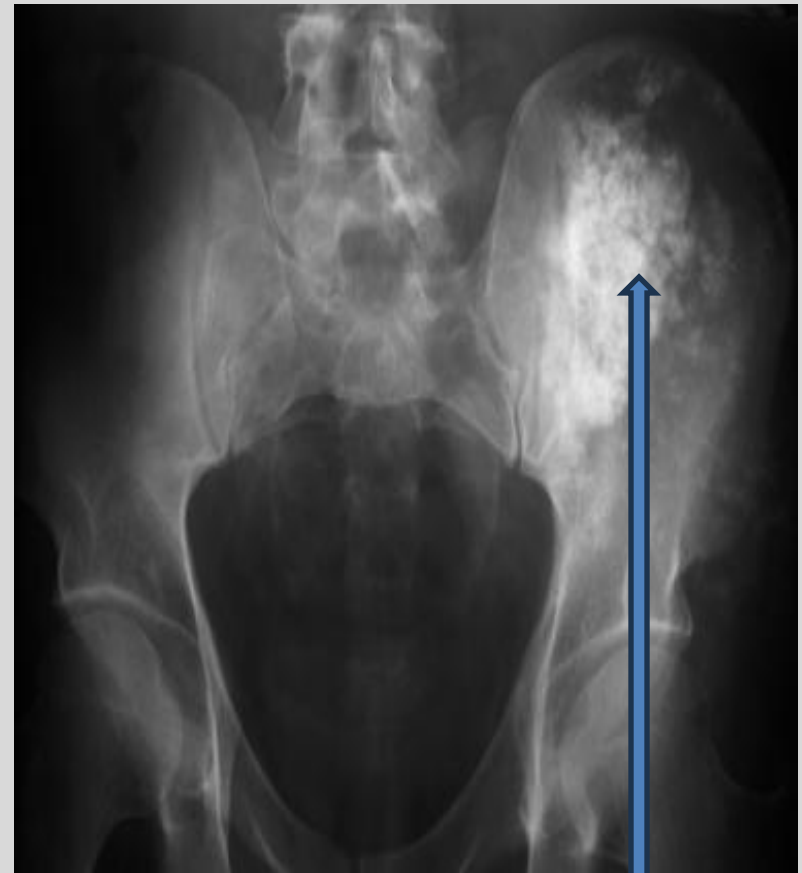
CHONDROSARCOMAS

- Painful, progressively enlarging masses
- Malignant cartilage-producing tumors;

Classification:

- Conventional
 - Clear cell
 - Dedifferentiated
 - Mesenchymal variants
-
- Patients: 40 years of age or older; men 2X more than women

Location: axial skeleton, especially pelvis, shoulder, and ribs



Xray - appears as foci of calcified flocculent densities

CONVENTIONAL CHONDROSARCOMA

Bulky tumors:

- nodules of glistening gray-white, translucent cartilage,
- matrix is often gelatinous or myxoid,
- ooze from cut surface;
- spotty calcifications typically present;
- central necrosis may create cystic spaces

Histo:

- neoplastic cartilage infiltrates marrow space
- variable cellularity, pleomorphism,
- degree of cytologic atypia
- mitotic activity;

Grading from 1 to 3; **Histologic grade correlates directly with biologic behavior**

Most conventional chondrosarcomas are grade 1 tumors

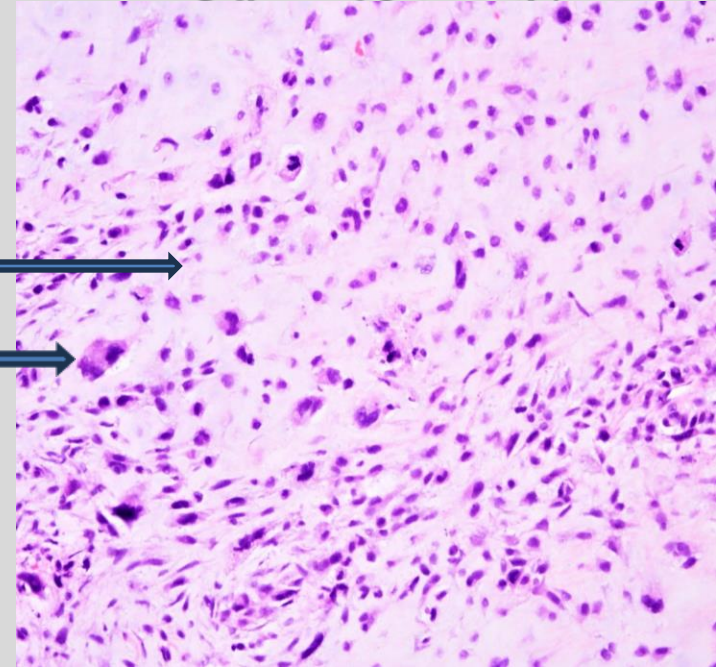
Rarely metastasize

5-year survival rates of 80% to 90%

70% of grade 3 tumors spread hematogenously, especially to lungs and the 5-year survival is 43%

Treatment: Conventional wide surgical excision; mesenchymal and dedifferentiated tumors (more aggressive): excision and adjuvant chemotherapy

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METASTATIC BONE TUMORS

Metastatic Cancers > Primary bone tumors

Mechanisms:

- direct extension
- lymphatic or hematogenous dissemination
- intraspinal seeding

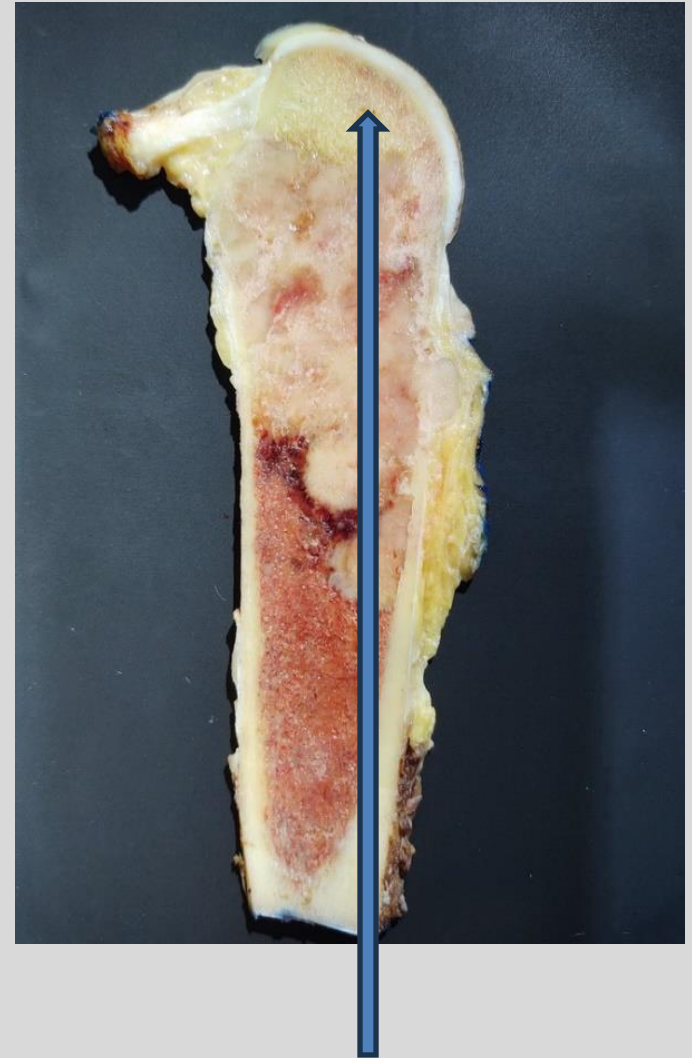
Adults: > 75% of skeletal metastases from prostate, breast, kidney, and lung cancers

Children: neuroblastoma, Wilms tumor, osteosarcoma, Ewing sarcoma, and rhabdomyosarcoma

- Usually multifocal

Xray: may be lytic (bone destroying), blastic (bone forming), or mixed lytic and blastic (prostatic adenocarcinoma is predominantly blastic; carcinomas of kidney, lung, and gastrointestinal tract and malignant melanoma are lytic)

Breast Cancer metastasizing to femur



Extension through the cortex into anterior soft tissue.

METASTATIC BONE TUMORS

Bidirectional interactions between metastatic cancer cells and native bone cells cause changes in bone matrix

Tumor cells do not directly resorb bone in lytic lesions, but secrete, prostaglandins, cytokines, and PTHrP



up-regulate RANKL on osteoblasts and stromal cells



stimulating osteoclast activity

Tumor cell growth supported by release of matrix-bound TGF- β , IGF-1, and FGF as bone is resorbed

Presence of bone metastases: poor prognosis

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Treatment - symptomatic relief of pain by analgesics and radiotherapy.

Prevention of further development of bone metastases by bisphosphonates
zoledronic acid

Lecture Feedback Form:

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